

Introduction to Foundry & AIP for Enterprise Organizations

Palantir Foundry & AIP 概念基础课程 · 按左侧导航顺序完整收录（23 个 lesson，含图片）

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课程总览

课程名称: Introduction to Foundry & AIP for Enterprise Organizations

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Course Overview

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Welcome to this course on the conceptual foundations of Foundry and AIP. This course is designed to give you a foundational understanding of the platform — its mission, its architecture, and its approach to turning enterprise data into operational decisions. Whether you're a data engineer, an operator on the plant floor, or a business leader charting your organization's AI strategy, this course will walk you through the key concepts that make Foundry and AIP a uniquely powerful system for operationalizing data and AI across your organization.

Over the next sections, you'll learn the "what and why" of Foundry and AIP. You'll understand how the platform is structured, what makes it different from traditional data platforms, and how its components work together to close the loop between data, decisions, and actions. After this course, you'll be well-positioned to start hands-on training.

The Enterprise Autonomy Vision

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The ambition behind Foundry and AIP is not to do what's minimally viable — it's to do what's maximally ambitious. The vision is an enterprise where thousands of AI agents wield thousands of specific tools across every workflow imaginable — not as a far-off aspiration, but as a concrete engineering target that shapes how the platform is built today.

Foundry and AIP are already deployed in some of the most demanding operational contexts in the world: staff scheduling in hospitals, underwriting workflows in insurance, build operations in shipyards, and defense and intelligence missions across the spectrum. The stakes are real, the data is messy, and the margin for error is thin.

This is what Palantir's forward-deployed approach to product development means in practice: working backwards from the hardest problems customers face, and letting those problems define what needs to be built. Above all else, it's a radical orientation around the outcome.

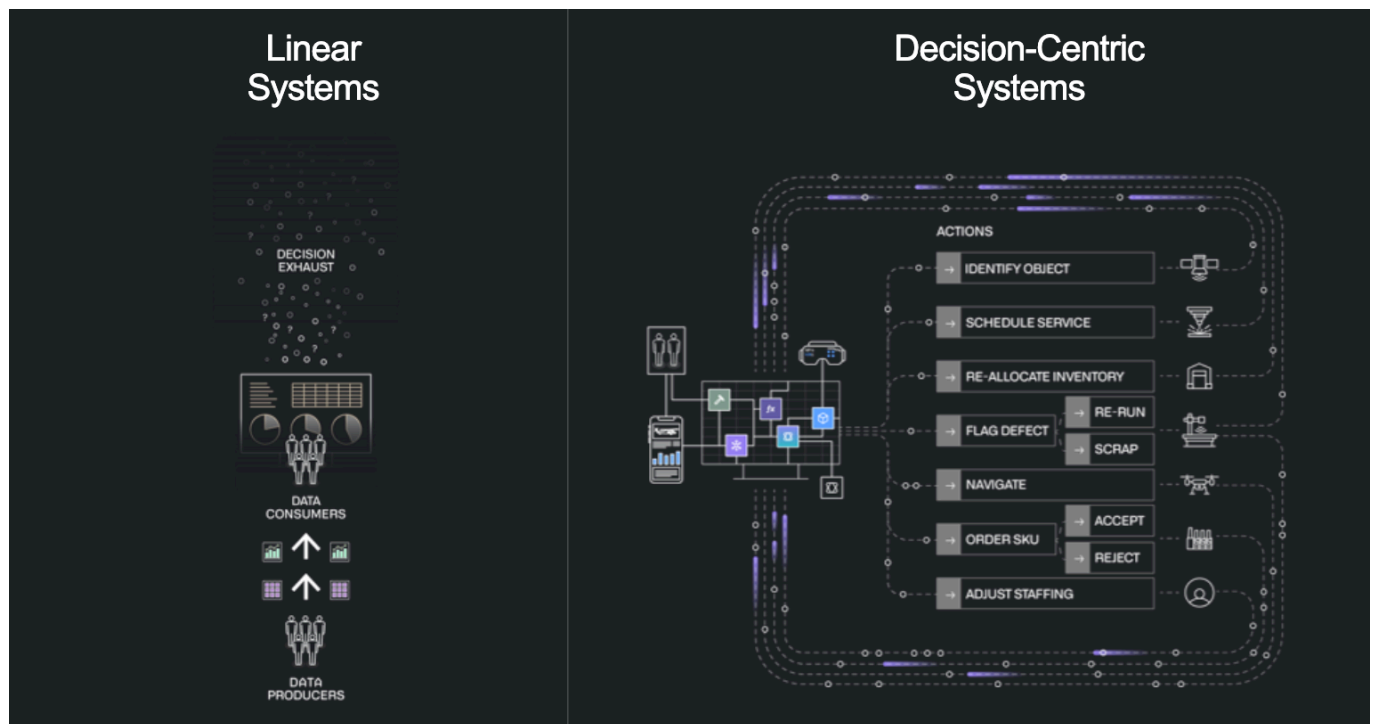
The Evolution from Linear to Decision-Centric Systems

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To understand why Foundry and AIP are architected the way they are, it helps to understand the problem they were built to solve.

Most organizations manage data like an assembly line: data producers collect it, integration teams combine it, transformation teams convert it, reporting teams generate dashboards, and decision-makers consume the results. This linear flow works — up to a point. But it leaves critical gaps. Learnings from data processing are rarely captured back into the system. Decisions made from insights aren't always materialized into action — someone sees a dashboard, realizes a shipment should be rerouted, but has to swivel-chair into a completely different system to actually make it happen. And there's often no effective feedback loop between the front-line operators who see what's really going on and the analytics teams who produce the reports.

Foundry and AIP are built to close these gaps: not just a data platform, but a decision-centric operating system where data flows in, decisions emerge, actions are taken, and the results feed back to improve future decisions.



AI Labor: A New Category

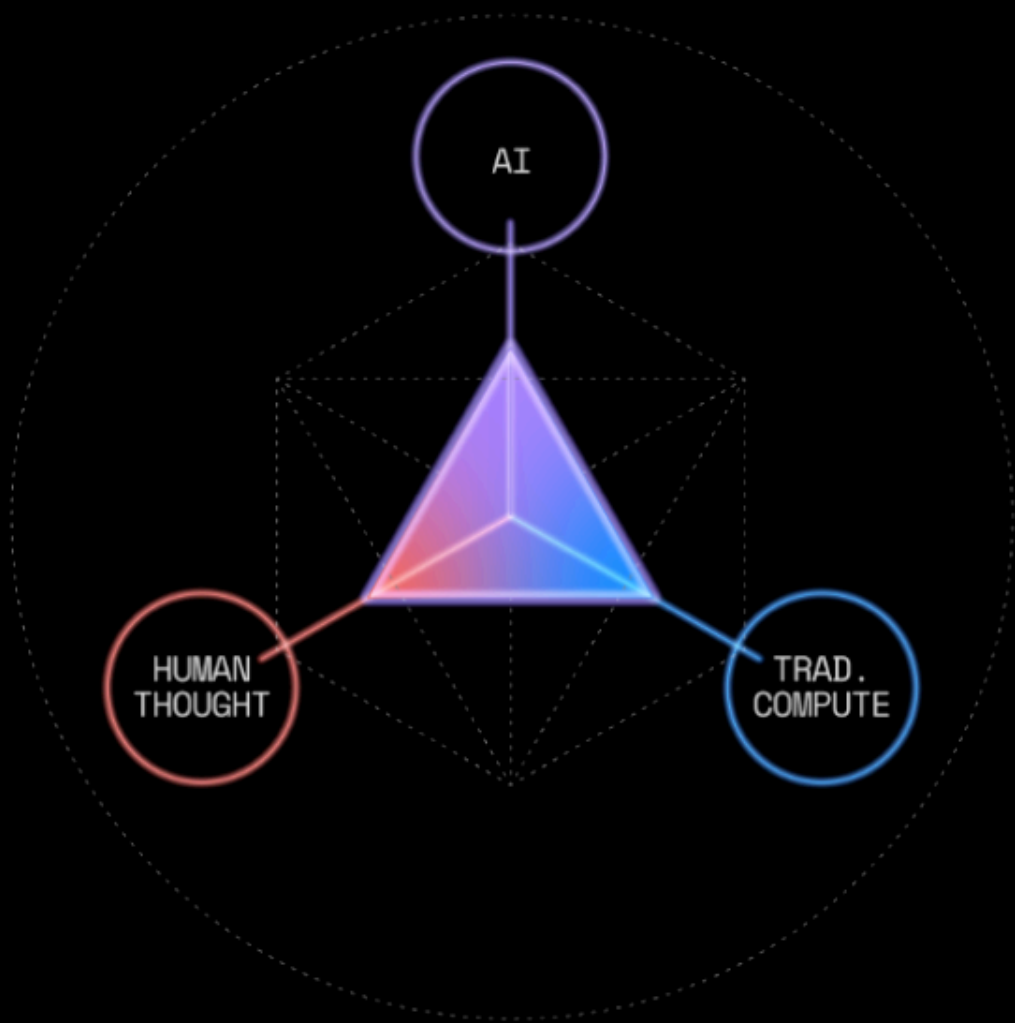
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Closing the loop from data to decisions to actions is hard enough with just people and software. But there's a reason this challenge is more urgent: AI has introduced a fundamentally new category of capability into the enterprise.

Traditionally, there are two categories of effort in any organization: human effort and thought, and traditional computation — databases, rule engines, scheduled jobs. AI introduces a third: what we can think of as AI Labor. It's not simply faster computation, and it's not a replacement for human judgment. It's a distinct category that sits as a conceptual bridge between the two.

What this means in practice is that building AI-powered operations requires what is essentially a rigorous form of alchemy: figuring out the right balance between human effort, AI labor, and traditional computation for each step of a workflow. And that balance isn't static — it shifts over time as you learn what works, as you equip agents with better tools and more context, and as you build confidence in the system's reliability.

This is the design principle behind the platform: Foundry and AIP don't just give you a place to store data or run models. They give you an architecture where human effort, AI labor, and traditional computation can be composed, recomposed, and progressively automated — all on a shared foundation — Ontology. For the decision-centric systems.



High-Level Architecture of the Platform

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So what does it actually take to deliver on this **ambition** — thousands of agents wielding thousands of tools, with the balance between human effort, AI labor, and traditional computation shifting smoothly across every workflow, for every decision?

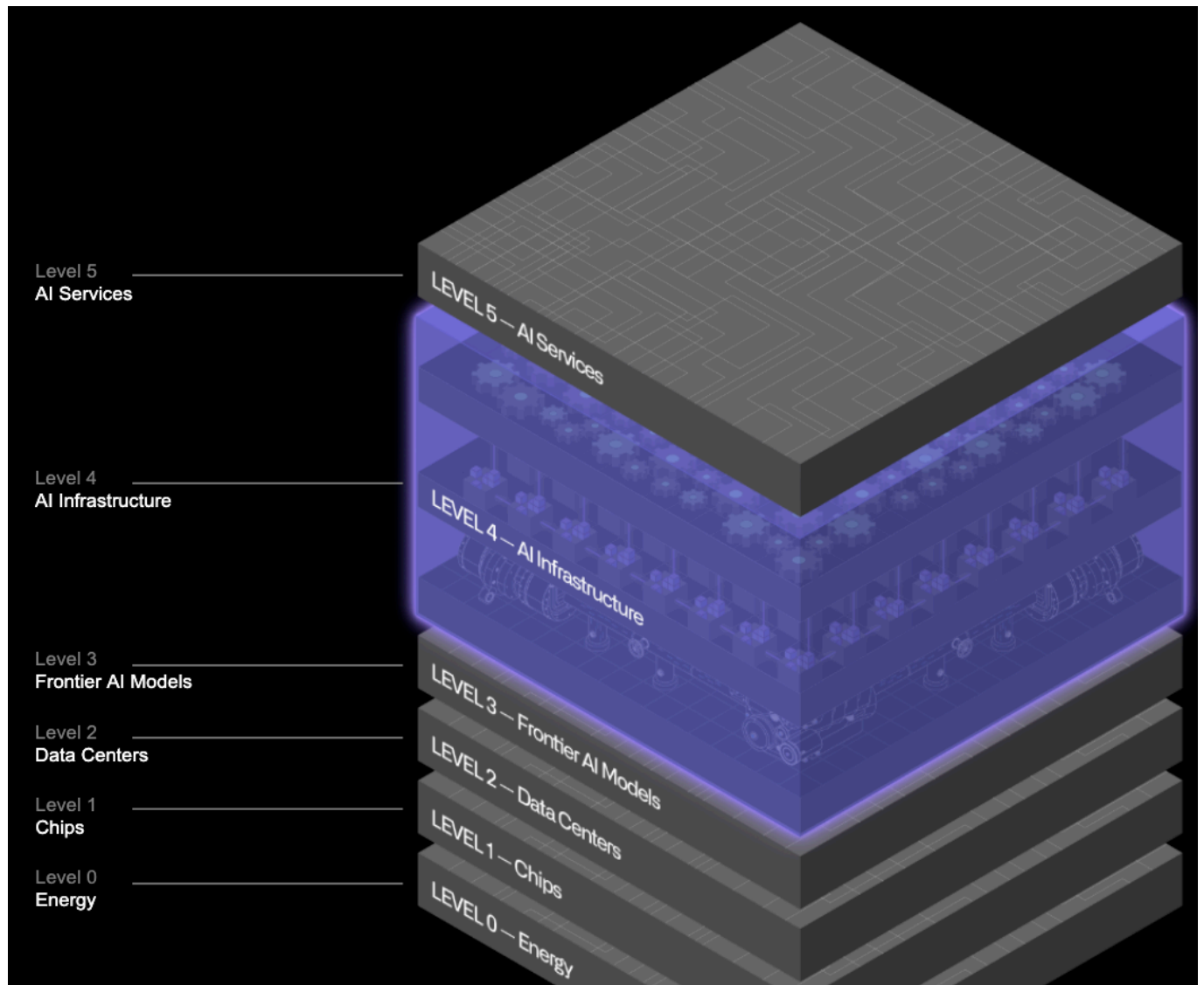
It takes more than a data lake. More than an ML platform. More than an application builder. It takes a unified architecture purpose-built for this kind of composability — and that architecture is organized around one central concept: the **Ontology**.

The Ontology is the layer at the heart of the platform. Ontology is the shared world where both humans and AI operate. Around the Ontology, the platform provides the full set of services needed to make this work at enterprise scale.



Taken together, this is what we call **AI Infrastructure** — the layer between raw AI capabilities and actual enterprise outcomes. Frontier models are increasingly powerful, but **a model alone can't understand your business, respect your governance policies, integrate with your existing systems, or take actions that**

affect real-world operations. AI Infrastructure is what bridges that gap. And at its center, the Ontology is what makes the whole thing composable.



The rest of this course walks through each of these layers in detail — starting with the Ontology at the center, then exploring what feeds into it and what's built on top of it.

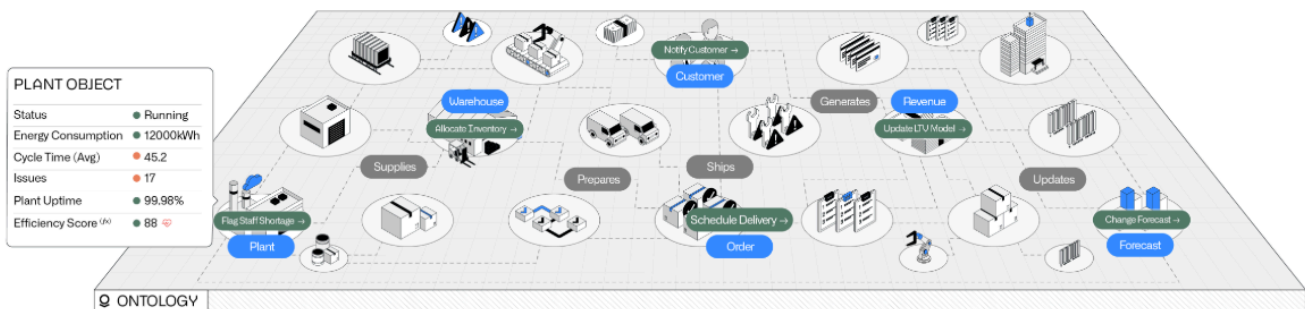
What is the Ontology?

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If you remember one thing from this course, let it be this: the Ontology is the centerpiece of the entire platform.

The Ontology is the nouns and verbs that make up your business. In a manufacturing context, it's plants and warehouses, suppliers and customers, shipments and orders. In healthcare, it's patients and providers, appointments and prescriptions, facilities and equipment. Whatever your domain, the Ontology models how your business actually operates — not how your IT systems need it to be structured so they can function, but how the real world actually works.

This is a semantic layer that bridges the gap between raw data from source systems and the business language that people use every day. It translates complex, multi-system data into familiar business terms, making the data accessible and actionable for everyone — from the data engineer who built the pipeline to the plant manager who needs to make a staffing decision in the next ten minutes.



The Ontology as the Shared World for Human + AI Teams

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This concept of "shared world" is crucial, and it's what makes the Ontology different from a traditional data warehouse or semantic layer.

For AI to be useful in enterprise settings, it needs more than data — it needs context. LLMs were not trained on your business's data or processes. The Ontology provides that context. When you build on the Ontology, you create a shared foundation that serves both humans and AI simultaneously.

The Ontology serves different audiences simultaneously, and this is one of its most powerful properties:

- **For humans**, it translates complex multi-system data into familiar business terminology. A plant manager doesn't need to know which SAP table holds the inventory data — they see "Warehouse X" with "42 days of supply" and can act on that directly.
- **For AI**, it provides the business context needed to reason effectively. When an LLM has access to the Ontology, it doesn't just see raw numbers — it understands that a warehouse is running low, that it's connected to specific plants, that there's a lead time constraint, and that certain actions are available to resolve the issue.
- **For systems**, it provides a single source of truth that eliminates the fragmentation caused by dozens of disconnected enterprise applications.
- **For governance**, it provides clear visibility into what data exists, who has access to it, what actions are being taken, and by whom (or what).

The Three Pillars of Decision-Centric Systems for Enterprise Automation

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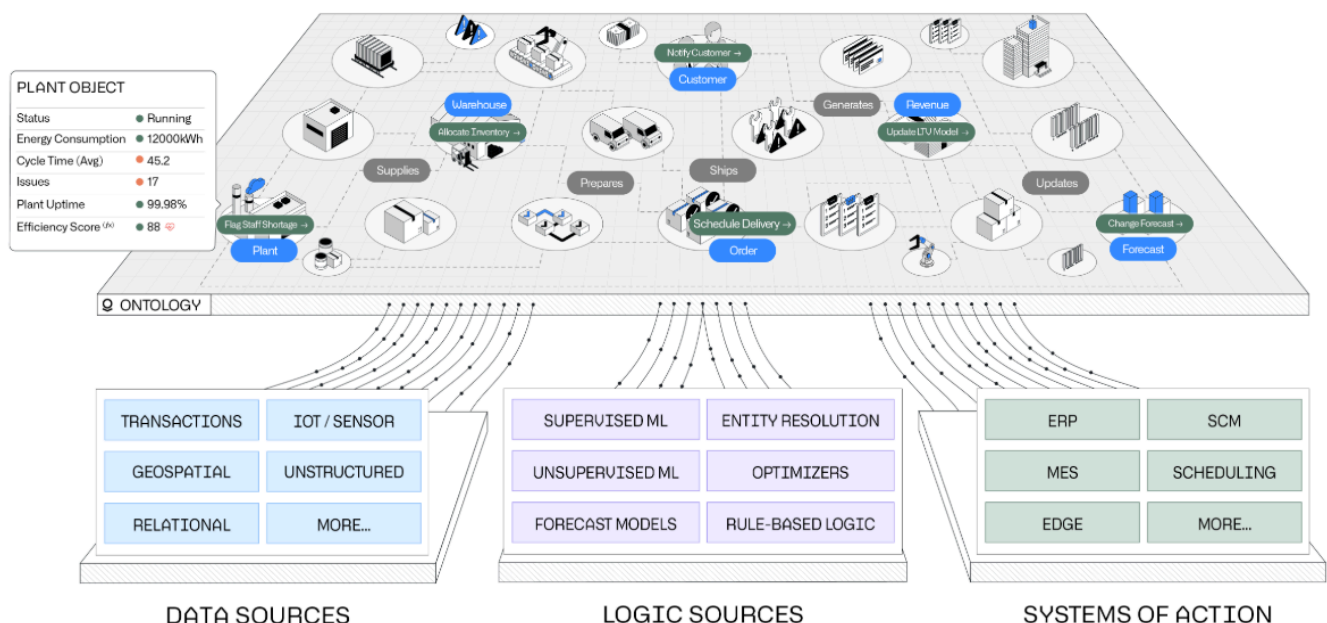
To make decisions in any business, you need three things: the data (what IS), the logic (how to THINK about it), and the actions (what you can DO). The Ontology brings all three together, enabling human and AI teams to leverage them.

When you build on the Ontology, you create a **shared foundation** that serves both humans and AI simultaneously:

- **Data** — all modalities of data (structured, unstructured, time series, geospatial) are unified into business objects that both people and AI agents can understand and query.
- **Logic** — business rules, ML models, optimizers, and all forms of traditional compute are modeled as part of the Ontology, so both human operators and AI agents can call upon the same reasoning tools.
- **Action** — the actions you can take to affect the real world (creating an order in SAP, adjusting a schedule, flagging an issue) are modeled in the Ontology with full governance guardrails, available to both humans and AI.

Finally, it is important to establish a **common security and governance framework** — this goes beyond simple role-based access controls to include purpose-based controls, data markings, classifications, and comprehensive audit capabilities that span both human and AI teams.

This is what enables composability. Because humans and AI are working in the same shared world — seeing the same data, reasoning with the same logic, taking actions through the same governed channels — you can smoothly dial up and down the distribution of labor between them.

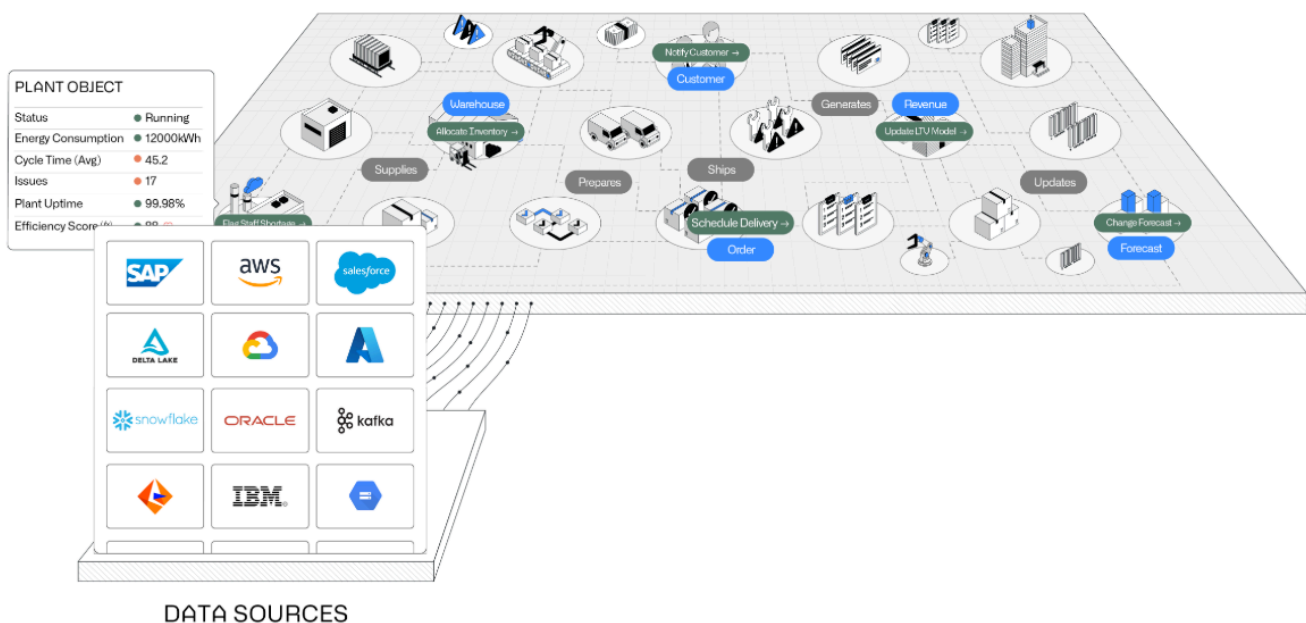


Now that we have a foundational understanding of the Ontology, the next step is to explore what lays under the Ontology and how it is populated. Next chapter will dive deeper into the mechanics and processes that support the Ontology.

2.1 Data Sources

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Foundry connects to the full breadth of enterprise data — regardless of where it lives, what format it's in, or how old the system is.



Types of data sources span the full spectrum of what enterprises generate: transactional data from ERPs and CRMs, IoT and sensor data from connected equipment, geospatial data from fleet tracking and logistics, unstructured data like documents, images, and maintenance logs, relational data from databases, and much more.

Native connectors provide out-of-the-box connectivity to over 300 enterprise systems. Whether you're pulling from SAP, Salesforce, ServiceNow, or a homegrown legacy system, Foundry has a connector for it. For custom or unusual sources, the platform provides frameworks for building your own connections.

The Multimodal Data Plane (MMDP) in practice is where the "any storage" promise becomes concrete. Through Virtual Tables, Foundry can work with data that lives in Snowflake, Databricks, BigQuery, or other platforms — without copying it. The data stays where it is. Foundry accesses it in place, through zero-copy federation. When transformations need to happen, pushdown compute sends the computation to where the data already lives, rather than moving the data to where the compute is. And all of this works with open data standards like Apache Iceberg, Parquet, and CSV, alongside standard interfaces like REST, JDBC, and S3-compatible APIs. To read more about Multimodal Data Plane, refer to [these public docs](#).

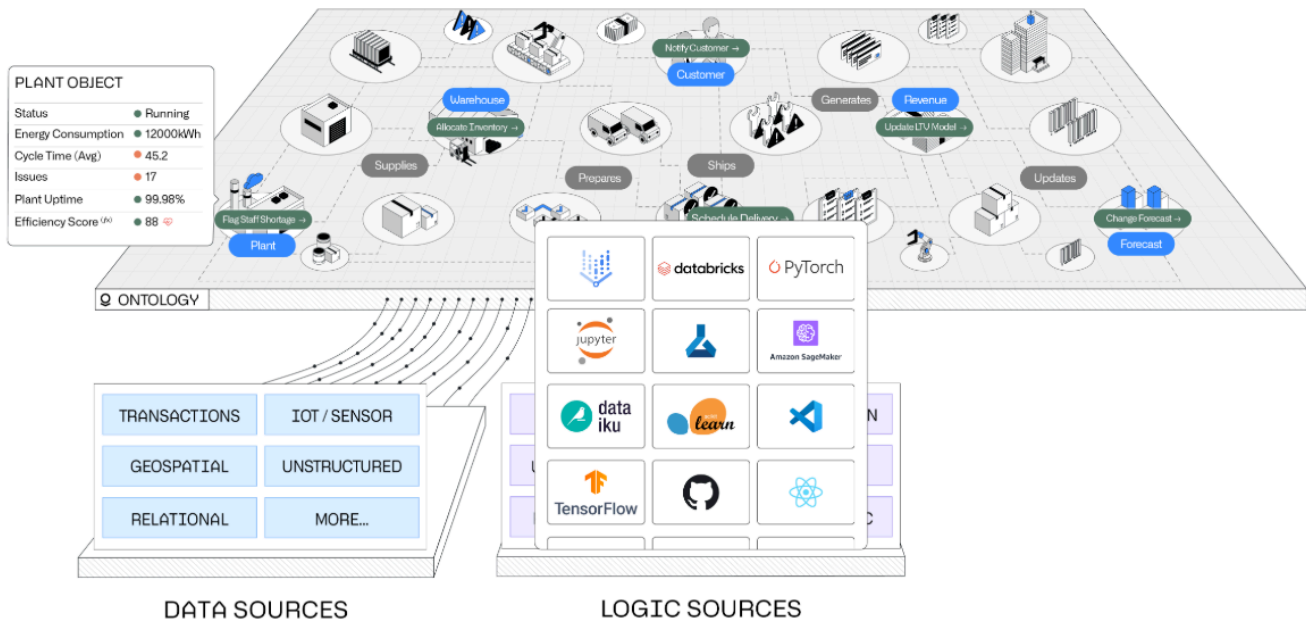
The critical insight here is that Foundry isn't asking you to abandon your data lake or warehouse — it's asking you to connect it. The Ontology layer above turns that connected data into context, and context is what makes

data operational.

2.2 Logic Sources

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Data tells you what IS. Logic tells you how to THINK about it.



Logic sources in Foundry span a wide range of complexity. At the simplest level, they might be rules captured in a spreadsheet, SOPs or tribal knowledge of your organization, all the way to sophisticated ML models for forecasting, third-party route optimizers, custom algorithms.

Integration tools support this full range. Models can be developed inside the platform or brought in from outside — containerized runtimes allow teams to package their existing model code in Docker containers and deploy them directly into the Foundry environment.

MMDP for compute extends the "any compute" philosophy to logic and transformation workloads. Foundry's compute-agnostic build framework supports mixing-and-matching of third-party compute runtimes.

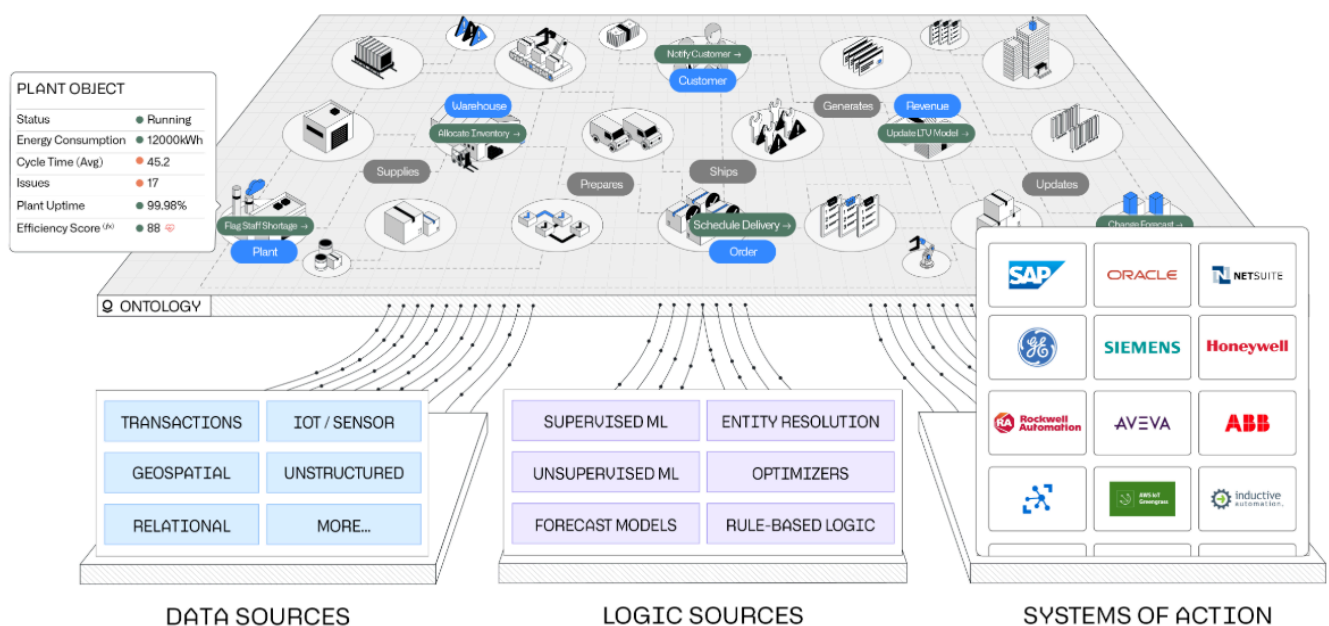
2.3 Systems of Action

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Data and logic are only valuable if they lead to action. Systems of Action represent the ability to write decisions back into the enterprise systems that run your operations.

If the you notice in the Ontology that Warehouse A is running low on a critical component and a forecasting model predicts that demand will spike next week, what do you do? You might create a stock transfer order, accelerate next shipment, or adjust the production schedule. Each of these is an action, and each one needs to land in the right enterprise system.

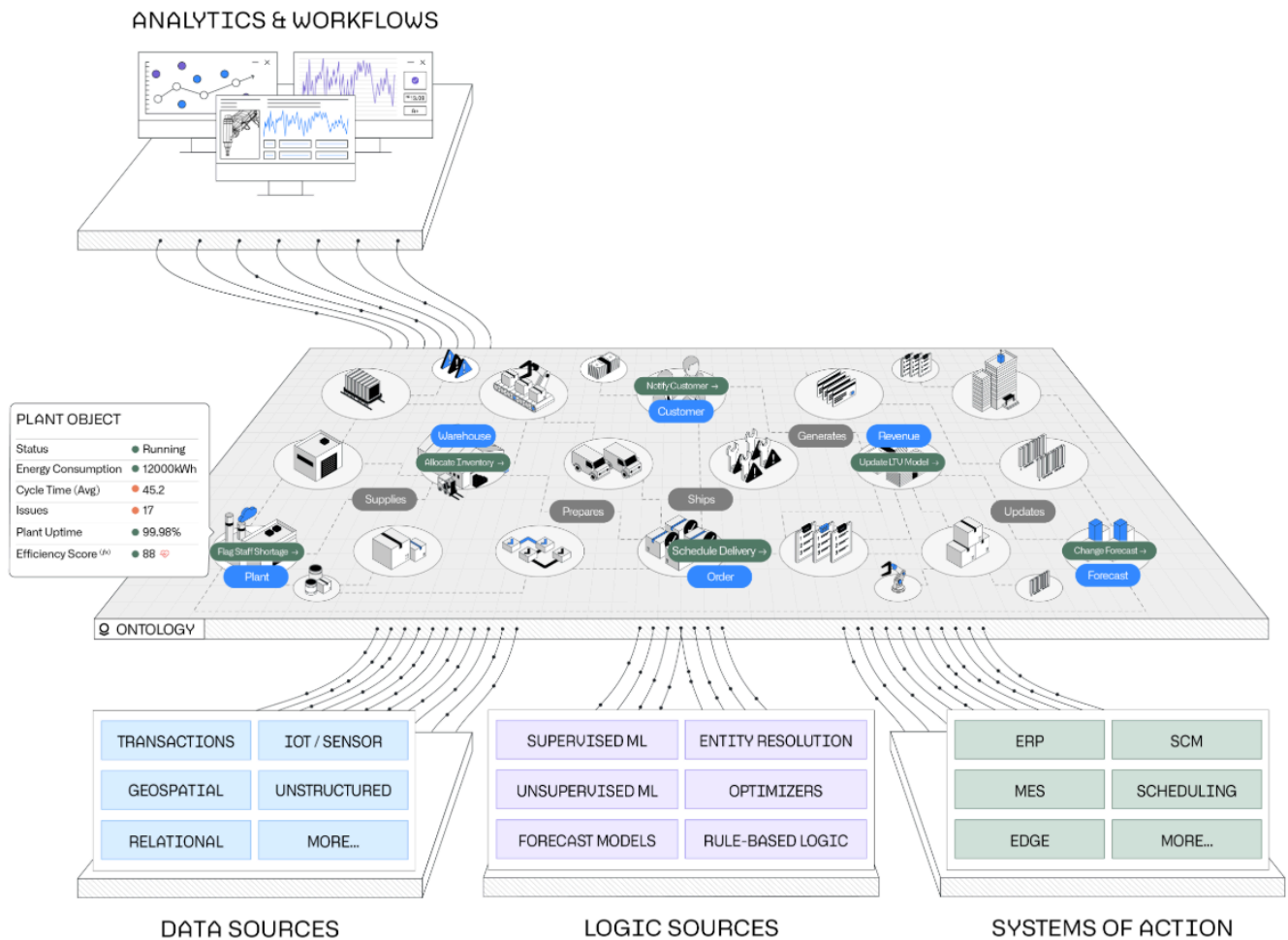
Foundry connects to all your systems of action. These actions are governed — permissions control who (or what) can execute them — and they're auditable, so you always know what happened and why.



3.1 Analytics & Workflows

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The first thing most organizations build on top of the Ontology is visibility. Analytics and workflows are where the Ontology becomes tangible — where people see their business reflected in data and start making decisions based on it.



Workflows connect people and processes across business functions. Rather than operating in silos, workflows orchestrate activity across the Ontology. A single workflow might span supplier monitoring, inventory allocation, production scheduling, and customer delivery, connecting the people responsible for each step and surfacing the information they need at each stage.

Real-time analytics and dashboards provide ongoing visibility into operations. Because they're built on the Ontology, these aren't static reports generated from a data warehouse — they reflect the live state of your business objects, complete with the logic and computed properties that make the data meaningful. Foundry provides a comprehensive workbench for building dashboards, reports, and visualizations, supporting low-code, no-code, and pro-code approaches.

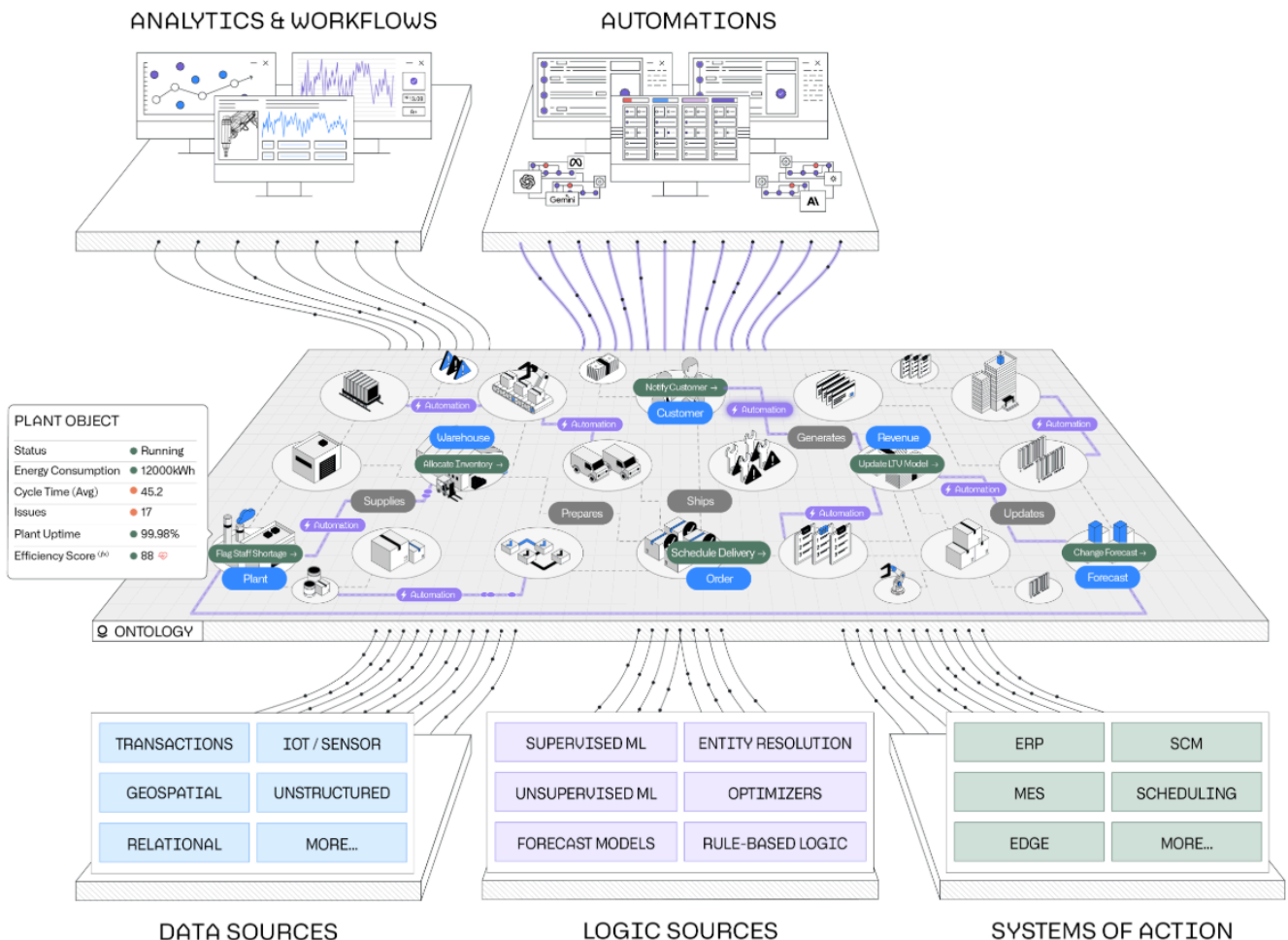
The workflow-analytics connection is important. In traditional BI, analytics are an end in themselves — you

produce a report and hope someone acts on it. In Foundry, analytics are a byproduct of the workflows you're building. The insights emerge from the operational processes, and decisions can be acted on immediately within the same environment.

3.2 Automations & Progressive Automation

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Automations in Foundry combine generative AI with the full context of the Ontology to create intelligent, governed workflows that can reason about your business and act on it.



Reasoning with Business Logic. One of the most powerful patterns in AIP is the ability for LLMs to call deterministic models as part of their reasoning chain. An LLM might interpret an unstructured maintenance report, and then call a deterministic failure-prediction model to assess the risk score — combining the LLM's ability to parse natural language with the precision of a deterministic tool. The LLM has access to both the crucial data and the business logic that are modeled in the Ontology. This is a critical distinction: the LLM isn't operating alone. It's orchestrating across multiple sources of logic — some AI-powered, some rule-based, some model-driven, some based on human intervention — all unified through the Ontology.

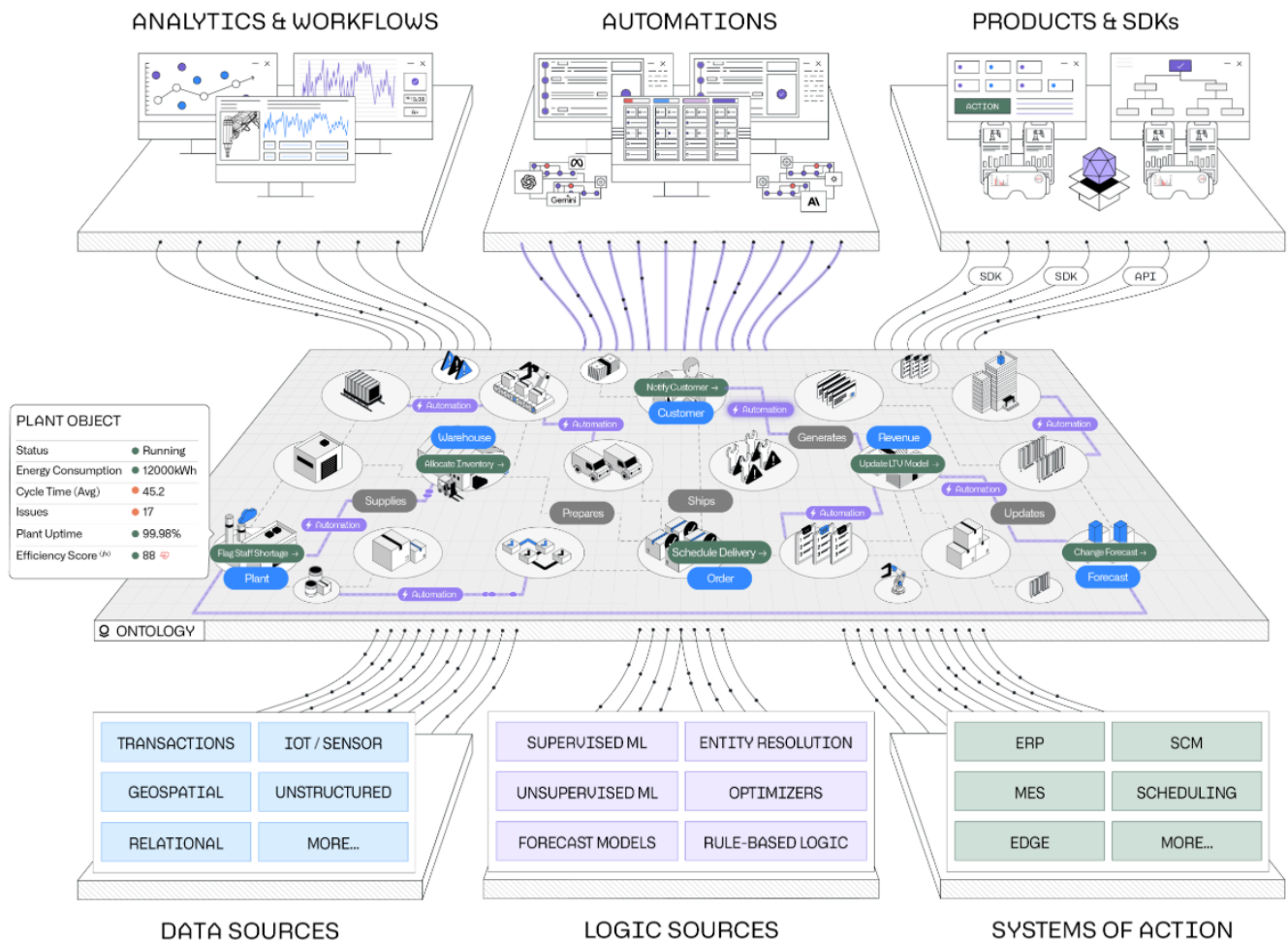
Orchestrating Complex Actions. Once you've captured a decision-making process in the Ontology, you can enhance it with agent orchestration. The agent isn't a standalone tool sitting outside your workflow. It's embedded directly into the business process, reasoning with the same logic and taking actions through the same governed channels that a human operator would use.

AIP Evals: Building Confidence for Agentic Workflows. Evals are the cornerstone of effective agentic workflows — because when agents reason with your Ontology, take actions in your systems, and operate inside your business processes, observability isn't optional. It's the foundation that makes everything else trustworthy. The platform provides a structured evaluation environment for testing AI-powered functions and agents, providing full observability over all workflows and automations.

3.3 Products & SDKs

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Finally, platform's capabilities can be packaged into products that reach end users — whether those are internal operators, external customers, or other systems.



The Ontology SDK (OSDK) is a generated software development kit that gives developers programmatic access to the Ontology from their own development environment. Developers can create custom React applications, Python backends, Java services, or integrations in any language via OpenAPI — all powered by the Ontology's data, logic, and actions.

In effect, the OSDK turns your Ontology into an SDK of your business. Agentic workflows built with the OSDK can be integrated into existing applications — your organization's existing tools don't need to be replaced; they can be enhanced with Ontology-powered capabilities.

Beyond APIs and SDKs, the platform supports integration through streaming and batch connectors with export plugins, webhooks for real-time write-back to external systems, and external transforms that allow data transformations to run outside of Foundry while integrating results back in.

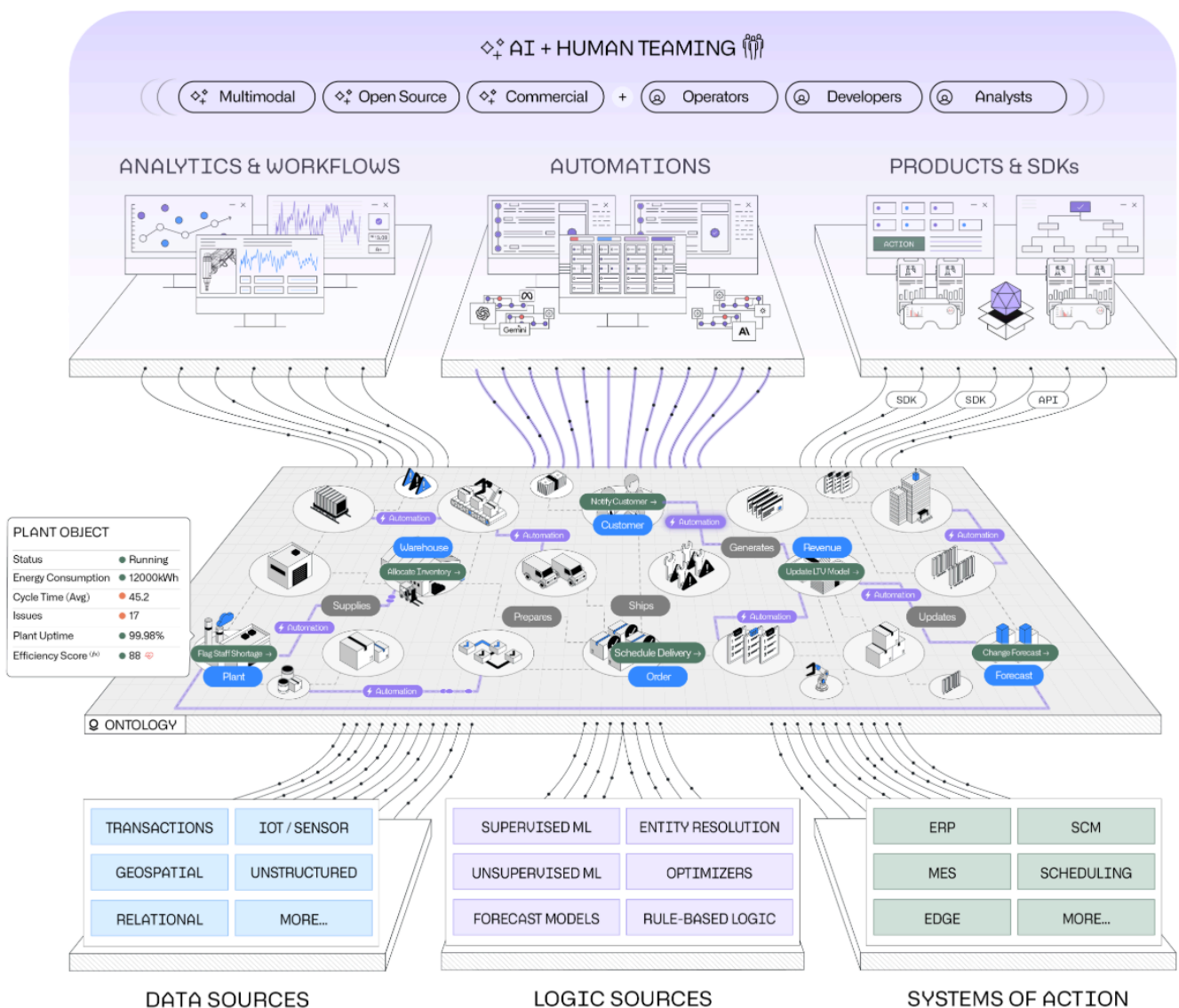
Common Foundation for Human + AI Collaboration

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This brings us to the core insight behind how the platform is actually used: the question is not human OR AI, but **human AND AI**, in the right composition. Every workflow involves a mix of human effort, AI labor, and traditional computation.

Tying it all together: the Ontology is what makes all of this possible. It's the shared world where humans see their business in familiar terms, AI agents access the context they need to reason effectively, every action is governed and auditable, and the composition of human and AI labor can shift over time without breaking the underlying architecture.

This is what composability means in practice. Because the Ontology provides a shared world where humans and AI can both operate — seeing the same data, using the same logic, taking actions through the same governed framework — you can adjust the mix fluidly.



The Role of Evals in Progressive Automation

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Each phase transition requires evidence. You can't move from human-led to AI-led on faith — you need to demonstrate that AI-generated recommendations match human-quality decisions at a sufficient rate, across a sufficient range of conditions.

This is where **AIP Evals** becomes structurally important. Evals provide the evidence base for expanding AI autonomy. Before each phase transition, teams run comprehensive evaluation suites that measure AI performance against defined criteria. As workflows mature, continuous monitoring and benchmarking ensure that performance doesn't degrade.

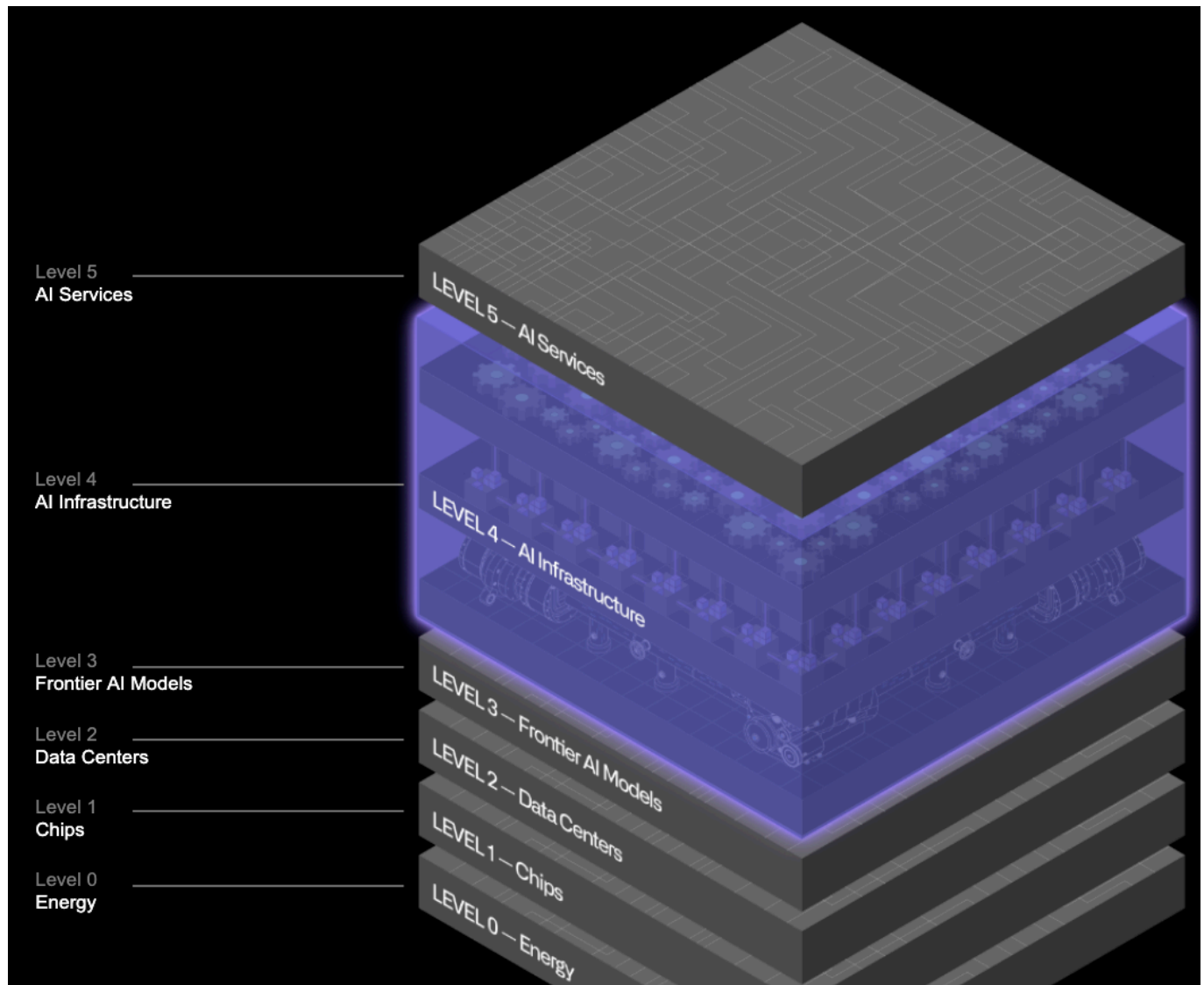
The combination of progressive automation as the strategy and Evals as the evidence mechanism creates a principled, defensible path from human-led operations to increasingly autonomous workflows — at whatever pace is appropriate for each specific use case.

From Frontier Models to Enterprise Outcomes

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A frontier model can generate text, reason about code, and analyze images. But an enterprise needs that model to understand its specific business context, access its proprietary data, respect its governance policies, integrate with its existing systems, and take actions that affect real-world operations. The gap between what frontier models provide and what enterprises need is enormous. Foundry and AIP bridge that gap.

If a frontier model is the brain of your Enterprise Agents, **Ontology is its body**.



Security & Governance Layer

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Everything described so far — the Ontology, the data and logic that feed it, the automations and products built on top of it — operates within a security and governance layer that spans the entire platform. This isn't a bolt-on. It's woven into every interaction, every data access, every action taken by a human or an AI agent.

Most enterprise platforms stop at role-based access control: user X has access to resource Y. Foundry goes significantly further.

- **Purpose-based controls** restrict data access not just by *who* is accessing it, but by *why*. A compliance analyst and a marketing analyst might both have access to customer records, but the permitted uses — and the data fields visible to each — can differ based on the purpose of their work.
- **Data markings and classifications** attach sensitivity labels that travel with the data through every transformation, every dashboard, every automation. Data classified as restricted at ingestion remains restricted when it surfaces in an application or gets passed to an LLM — the markings don't get left behind.
- **Active controls** enforce policies dynamically based on context, rather than relying solely on static permission grants.

In a world where AI agents are taking actions alongside human operators, audit trails become more important than ever. Foundry provides **comprehensive logging** for both human and AI activity. This isn't just for compliance — it's the foundation for the trust that enables progressive automation. You can't confidently expand AI autonomy if you can't see exactly what the AI did and why.

Branching: Safe Change Management for Human and AI Builders

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Finally, Foundry and AIP are platforms for builders.

When both humans and AI agents are building on the same platform, you need a mechanism that prevents changes from breaking production until they've been reviewed. This is what **Foundry's global branching** model provides, allowing for safe human + AI teaming in the builder world.

This isn't just version control. It's a governance mechanism — runtime controls govern what agents and users *can do* in production, branching governs what changes *make it to* production in the first place.

Apollo - Software Delivery Layer

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Apollo is Palantir's platform for autonomous software deployment and management. It's the engine that ensures everything described in this course can actually run — wherever it needs to.

The defining capability is deployment flexibility. Apollo supports public cloud environments (AWS, Azure, GCP), on-premises data centers, edge locations, hybrid configurations, and fully air-gapped environments with no internet connectivity. This matters because enterprises operate across all of these environments, often simultaneously, and AI infrastructure needs to meet them where they are.

Apollo handles continuous delivery, configuration management, and operational monitoring across all deployment environments, ensuring that the same platform capabilities are available regardless of where the software is running.

To learn more about Apollo, refer to these public [docs](#).

Course wrap-up

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You've now seen the full picture: data, logic, and actions flowing into the Ontology. Analytics, automations, and products built on top of it. Security and governance woven through all of it. Any storage, any compute, any model — no lock-in, no rip-and-replace.

At the center of it all, one idea: **Ontology as the shared world where humans and AI operate side by side** — seeing the same data, reasoning with the same logic, taking actions through the same governed channels. Start with humans in the lead. Layer in AI where it earns trust. Expand autonomy as the evidence builds.

Resources & Next Steps

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You can explore these additional resources as a next step to get hands-on training to the platform and keep learning about Foundry and AIP capabilities.

Suggested Further Training

- [Speedrun: Your First End-to-End Workflow](#)
- [Foundry & AIP Aware](#), incl. exam & certification
- Explore Training Tracks on the [Palantir Learn homepage](#)

Documentation and Support

- Visit the [Platform Documentation](#) to learn more
- Join the [Community](#) to ask question and share knowledge
- Follow [Palantir Developers YouTube Channel](#) for latest platform updates

Collection of Resource mentioned in the course

- [Palantir Foundry Ontology](#)
- [Interoperability I Multimodal Data Plane](#)
- Discover under-explored platform components like [Apollo](#)

We also encourage you to enquire internally about groups, pages or channels that may exist in your Organization to share Foundry and AIP best practices or internal resources.

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